

## SAFETY DATA SHEET

Creation Date 26-Sep-2009

Revision Date 02-Apr-2024

Revision Number 3

### 1. Identification

**Product Name** Cobalt(II) sulfate heptahydrate

**Cat No. :** C21310

**CAS No** 10026-24-1  
**Synonyms** Cobalt(II) sulfate (1:1) heptahydrate.; Cobalt sulfate heptahydrate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                           |             |
|---------------------------|-------------|
| Acute oral toxicity       | Category 4  |
| Respiratory Sensitization | Category 1  |
| Skin Sensitization        | Category 1  |
| Germ Cell Mutagenicity    | Category 2  |
| Carcinogenicity           | Category 1B |
| Reproductive Toxicity     | Category 1B |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Harmful if swallowed  
May cause an allergic skin reaction  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
Suspected of causing genetic defects  
May cause cancer by inhalation  
May damage fertility



### Precautionary Statements

#### Prevention

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
In case of inadequate ventilation wear respiratory protection  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves

#### Response

IF exposed or concerned: Get medical attention/advice

#### Inhalation

IF INHALED: If breathing is difficult, remove to fresh air and keep at rest in a position comfortable for breathing  
If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician

#### Skin

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse

#### Ingestion

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

#### Storage

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

Very toxic to aquatic life with long lasting effects  
WARNING. Cancer - <https://www.p65warnings.ca.gov/>.

## 3. Composition/Information on Ingredients

| Component                       | CAS No     | Weight % |
|---------------------------------|------------|----------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | <=100    |
| Cobalt sulfate                  | 10124-43-3 | -        |

## 4. First-aid measures

#### Eye Contact

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | clothes and shoes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                            |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                           | Clean mouth with water. Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                  |

## 5. Fire-fighting measures

|                                         |                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                     |
| <b>Flash Point</b>                      | No information available                                                     |
| <b>Method -</b>                         | No information available                                                     |
| <b>Autoignition Temperature</b>         | No information available                                                     |
| <b>Explosion Limits</b>                 |                                                                              |
| <b>Upper</b>                            | No data available                                                            |
| <b>Lower</b>                            | No data available                                                            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                     |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                     |

### Specific Hazards Arising from the Chemical

Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Sulfur oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 0                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                                                                                                 |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

|                 |                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Wash hands before breaks and |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

immediately after handling the product.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                       | ACGIH TLV                   | OSHA PEL | NIOSH | Mexico OEL (TWA)            |
|---------------------------------|-----------------------------|----------|-------|-----------------------------|
| Cobalt(II) sulfate heptahydrate | TWA: 0.02 mg/m <sup>3</sup> |          |       | TWA: 0.02 mg/m <sup>3</sup> |
| Cobalt sulfate                  | TWA: 0.02 mg/m <sup>3</sup> |          |       | TWA: 0.02 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Particulates filter conforming to EN 143.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Solid                    |
| Appearance                             | Red brown                |
| Odor                                   | Odorless                 |
| Odor Threshold                         | No information available |
| pH                                     | 4-5 (34 %)               |
| Melting Point/Range                    | 98 °C / 208.4 °F         |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Co O4 S . 7 H2 O         |
| Molecular Weight                       | 281.09                   |

## 10. Stability and reactivity

|                                         |                                                        |
|-----------------------------------------|--------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available             |
| <b>Stability</b>                        | Hygroscopic.                                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                |
| <b>Hazardous Decomposition Products</b> | Sulfur oxides                                          |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.               |
| <b>Hazardous Reactions</b>              | None under normal processing.                          |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component                       | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|---------------------------------|--------------------------|-------------|-----------------|
| Cobalt(II) sulfate heptahydrate | LD50 = 582 mg/kg ( Rat ) | Not listed  | Not listed      |
| Cobalt sulfate                  | LD50 = 424 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | May cause irritation                                                                     |
| <b>Sensitization</b>   | May cause sensitization by skin contact                                                  |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                       | CAS No     | IARC     | NTP                    | ACGIH | OSHA | Mexico     |
|---------------------------------|------------|----------|------------------------|-------|------|------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | Group 2B | Reasonably Anticipated | A3    | X    | Not listed |
| Cobalt sulfate                  | 10124-43-3 | Group 2B | Reasonably Anticipated | A3    | X    | Not listed |

*IARC (International Agency for Research on Cancer)*

*NTP: (National Toxicity Program)*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

**Mutagenic Effects** Mutagenic effects have occurred in experimental animals.

**Reproductive Effects** Experiments have shown reproductive toxicity effects on laboratory animals.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

|                                            |                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STOT - single exposure                     | None known                                                                                                                                                                            |
| STOT - repeated exposure                   | None known                                                                                                                                                                            |
| Aspiration hazard                          | No information available                                                                                                                                                              |
| Symptoms / effects, both acute and delayed | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| Endocrine Disruptor Information            | No information available                                                                                                                                                              |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                        |

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component      | Freshwater Algae                                              | Freshwater Fish | Microtox   | Water Flea |
|----------------|---------------------------------------------------------------|-----------------|------------|------------|
| Cobalt sulfate | EC50: 0.4 - 72 mg/L, 72h<br>(Pseudokirchneriella subcapitata) | Not listed      | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

### TDG

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

### IATA

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.* |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

### IMDG/IMO

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

## 15. Regulatory information

### United States of America Inventory

| Component                       | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | -    | -                                             | -                           |
| Cobalt sulfate                  | 10124-43-3 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                       | CAS No     | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|---------------------------------|------------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | -   | -   | -         | X     | X    | X    | X    | X     | -        |
| Cobalt sulfate                  | 10124-43-3 | X   | -   | 233-334-2 | X     | X    | X    | X    | X     | KE-06128 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component                       | CAS No     | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------------------------|------------|----------|-------------------------------|---------------------------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | <=100    | 0.1 %                         | -                               |
| Cobalt sulfate                  | 10124-43-3 | -        | 0.1 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

| Component                       | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|---------------------------------|-----------|-------------------------|-------------------------|
| Cobalt(II) sulfate heptahydrate | X         |                         | -                       |
| Cobalt sulfate                  | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component                       | CAS No     | California Prop. 65 | Prop 65 NSRL | Category   |
|---------------------------------|------------|---------------------|--------------|------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | Carcinogen          | -            | Carcinogen |
| Cobalt sulfate                  | 10124-43-3 | Carcinogen          | -            | Carcinogen |

**U.S. State Right-to-Know Regulations**

| Component                       | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------------------|---------------|------------|--------------|----------|--------------|
| Cobalt(II) sulfate heptahydrate | -             | X          | X            | X        | -            |
| Cobalt sulfate                  | -             | X          | X            | X        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                       | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                  | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------|------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | -                                                                   | -                                                                                                                                                                                                              | -                                                                                                     |
| Cobalt sulfate                  | 10124-43-3 | -                                                                   | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 233-334-2 - Carcinogenic, Article 57a; Toxic for reproduction, Article 57c      |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

**REACH links**
<https://echa.europa.eu/authorisation-list>
<https://echa.europa.eu/substances-restricted-under-reach>
<https://echa.europa.eu/candidate-list-table>
**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                       | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |



|                |            |        |                |                |                |
|----------------|------------|--------|----------------|----------------|----------------|
| Cobalt sulfate | 10124-43-3 | Listed | Not applicable | Not applicable | Not applicable |
|----------------|------------|--------|----------------|----------------|----------------|

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                       | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cobalt(II) sulfate heptahydrate | 10026-24-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Cobalt sulfate                  | 10124-43-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By** Health, Safety and Environmental Department  
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www.thermofisher.com

**Creation Date** 26-Sep-2009  
**Revision Date** 02-Apr-2024  
**Print Date** 02-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**